



A study of the primary crystallization of a mixed anion silicate glass

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ABSTRACT

This paper presents the results of a study of the primary crystallization of a multicomponent mixed anion silicate glass. The primary phase, leucite, and the secondary phase, diopside, were formed by surface crystallization, while the secondary phase, phlogopite, was formed by volume crystallization. The influence of the particle size of glass powder samples in the range 0–1 mm on the temperature of the DTA crystallization peak, T_p , the height of the peak, $(\delta T)_p$, and the parameter $T_p^2/(\Delta T)_p$ was studied. The behaviors of the parameters $(\delta T)_p$ and $T_p^2/(\Delta T)_p$ depend only on the change in the surface-to-volume nuclei ratio, as is the case with polymorphic crystallization. However, the particle size ranges in which the surface, volume and mixed crystallization mechanism dominate are considerably narrower for this glass. The influence of the duration of a pre-DTA heat treatment on the parameters T_p , $(\delta T)_p$, and ΔT_p for fixed temperatures in the range $T = 550$ – 750 °C was investigated. The T_p of the samples thermally treated for different times, at fixed temperatures, decreased up to $t = 5$ h and then increased to a value close to that of an as-quenched sample, after which the value remained constant. The curves of $(\delta T)_p$ and ΔT_p as a function of time for fixed temperatures show a maximum. The influence of the temperature of the pre-DTA heat treatment on the parameters T_p , $(\delta T)_p$, and ΔT_p for fixed times of $t = 1$ – 5 h was also investigated. On increasing the pre-DTA heat treatment time, the curves changed significantly. The curves for 3 and 5 h in the temperature range 580–660 °C were similar to the nucleation curve, which indicated that the volume nucleation process proceeded in this temperature range. The behavior of these parameters, as a result of the simultaneous action of different nucleation mechanisms and crystal growth differ from those previously reported for the case of polymorphic crystallization.

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1. Introduction

Recent studies have shown that only some glass compositions undergo volume nucleation. Mostly, these compositions crystallize polymorphically (i.e., the compositions of the crystal phase and glass are the same). For most other glasses, an efficient method to induce volume nucleation is the addition of a nucleating agent to the base composition. The role of these agents is complex [1,2]. It may be attributed to the influence of the ions which are introduced into the glass. They change the properties of the glass and, consequently, the nucleation changes from a heterogeneous to a homogeneous one. However, in addition to the type of ions introduced into a glass by the nucleating agent, its quantity is very important because in such a way a new multi-component composition is formed and during crystallization, beside the primary crystalline phase, secondary ones also appear. These compositions crystallize mostly by primary crystallization (i.e., the composition of the crystal phases and the glass are different) and, consequently,

in addition to the primary crystal phases, secondary ones are formed. Both nucleation mechanisms can act simultaneously, exerting an influence on the properties of newly formed material. Therefore, the selection of the type of nucleating agent and its quantity is not simple; hence a detailed study is necessary.

In this study, the base glass composition: 53.5 SiO₂ · 11.5 Al₂O₃ · 12.5 CaO · 16.5 MgO · 6.5 K₂O (mol%) was chosen for investigation. Recent studies [3,4] showed that during crystallization of a glass of similar composition, leucite was formed as the primary phase. Leucite is a framework pseudotetragonal aluminosilicate. Its structure consists of corner-linked (Al,Si)O₄ tetrahedra filled with K⁺ cations. Leucite is usually formed during surface crystallization of aluminosilicate glasses. The mixed anion silicate glass containing 2 mol% of F[−] employed in this study was obtained by substituting CaO by CaF₂ in the initial glass composition. The presence of F[−] ions should enable the formation of a secondary phase during volume crystallization. The aim of the current work was to study the crystallization behavior of a glass in which both nucleation mechanisms act simultaneously, heterogeneous nucleation of the primary phase and homogeneous nucleation of the secondary one. The experiments were performed under isothermal and non-isothermal conditions.

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2. Experimental

The raw materials used were reagent grade Al_2O_3 , MgO , CaO , CaF_2 and K_2CO_3 , and high-purity quartz (>99.5% SiO_2). The appropriate batch composition was melted at 1530 °C for 1 h in a Pt crucible. The melts were cast on a steel plate and cooled in air. Powder X-ray diffraction analysis (XRD) confirmed the quenched melts were vitreous. The glass samples were transparent, without visible residual gas bubbles.

The experiments under isothermal conditions were performed with bulk samples, in a one and two-stage regime in electric furnaces, Carbolite CWF 13/13 and BLF 17/3, with automatic regulation and a temperature accuracy in the range of ± 1 °C. In two-stage regime the samples were first treated isothermally at the nucleation temperature between $T = 590$ – 700 °C for different times, followed by further heating until the crystallization temperature was attained. The crystallization temperatures were in the range $T_c = 900$ – 1485 °C. The samples were kept at the crystallization temperatures for different times, in the range $t_c = 0.5$ – 20 h. Finally, the samples were removed from the furnace and crushed in an agate mortar. Powdered samples were used for X-ray analysis and fractured ones for SEM analysis.

A Jeol JSM 840A microscope was used for the SEM investigations. The samples were gold sputtered in a Jeol JFC 1100 ion sputter.

The XRD method was used to determine the phase composition. The XRD patterns were obtained on a Philips PW-1710 automated diffractometer using a Cu tube operated at 40 kV and 32 mA. The instrument was equipped with a diffracted beam curved graphite monochromator and a Xe-filled proportional counter. The diffraction data were collected in the 2θ Bragg angle range from 4 to 70°, counting for 0.25 s (qualitative identification) and 4 s (calculation of the unit-cell parameters) at every 0.02° step. The divergence and receiving slits were fixed 1 and 0.1, respectively. All the XRD measurements were performed at room temperature in a stationary sample holder. The LSUCRIPC program was used for the refinement of the cell dimensions from the powder data [5]. The quantitative amounts of the crystalline phases in the sample were determined using the full structure matching mode of the Rietveld refinement technique [6], using the FULLPROF program [7].

Differential thermal analysis, DTA, experiments were performed under non-isothermal conditions using a Netzch STA 409 EP instrument and Al_2O_3 powder as the reference material. Powder samples of the following granulations were prepared: <0.038; 0.038–0.053; 0.053–0.1; 0.1–0.212; 0.212–0.3; 0.3–0.4; 0.4–0.63; 0.63–0.7 and 0.7–1.0 mm. In the experiments, a constant weight (100 mg) of the samples was heated at a heating rate $v = 10$ °C/min. Also, the nucleation process was studied using powder samples with 0.4–0.63 mm granulation. These samples were heated at a rate of 10 °C/min up to a temperature in the range $T = 550$ – 750 °C and kept at the chosen temperature for different times; subsequently, the samples were heated at the same rate to $T = 1100$ °C.

Dilatometry measurements were performed using a BÄHR 8025 instrument on samples 3 mm in diameter and 40 mm long. The dilatometric testing procedure was performed in accordance with ISO 7884–8 [8].

3. Results

3.1. The chemical composition of the glass

The results of the chemical analyses of the sample designated G-3 are shown in Table 1. Also, for comparison, the initial composition is given. In the G-3 glass, the CaO content of 12 mol% was substituted by 4.2 mol% CaF_2 , 3.8 mol% MgO and 2.8 mol% K_2O . In such a way, the stable G-3 glass containing 2 mol% F^- anions and a high content of MgO was attained. Previous investigations of crystallization in aluminosilicate glasses containing fluorine and magnesium showed that CaF_2 segregates first if the content of MgO is low and mica type silicate segregates first if the content of F^- anions is >1 wt% and the MgO content is >5 wt% [9]. Therefore, it may be expected that the G-3 glass would enable the formation of a secondary phase during homogeneous nucleation.

3.2. Thermal properties of the glass

The DTA curve for the powder sample with particle sizes between 0.4 and 0.63 mm recorded at a heating rate of 10 °C/min up to the maximum operating temperature of the instrument, $T = 1300$ °C is shown in Fig. 1. One exothermic and two endothermic peaks were registered. The significant temperatures that were observed are T_g (677 °C), T_p (990 °C), T_{l-ph} (1158 °C), T_{l-D} (1224 °C). The T_g temperature on the DTA curve well matched the dilatometric measurement, i.e., a glass transition temperature of $T_g = 680$ °C and a dilatometric softening temperature of $T_{om} = 756$ °C were determined. The results of the XRD analysis (Section 3.3) showed that the exothermic peak corresponds to the formation of three crystalline phases: leucite as primary, and phlogopite and diopside as secondary ones. The endothermic peak temperatures correspond to the T_{l-ph} -liquidus temperature of phlogopite and the T_{l-D} -liquidus temperature of diopside. The liquidus temperature of the primary phase leucite, $T_{l-L} = 1485$ °C, was determined as the lowest temperature at which the glass could be held for a prolonged period of time without the formation of a crystalline phase.

3.3. X-ray diffraction analysis

In order to identify the formed crystal phases, the bulk samples were crystallized in a one and two-stage heat treatment (Table 2.)

The first group of experiments was performed in a one step regime by heating the samples at $T = 590$ – 700 °C for times of $t = 1$ – 20 h. The XRD patterns of these samples did not show the separation of crystalline phases. The second group of experiments was performed in a two-step heating regime. In the first step, the samples were heated at nucleation temperatures of $T_n = 590$, 630 and 680 °C for a time $t_n = 20$ h. Then the heating was continued at a rate of $v = 10$ °C/min up to the crystallization temperatures $T_c = 900$ – 1100 °C and then held for times $t_c = 0.5$ – 20 h. The XRD patterns of these samples showed that at $T_c = 900$ °C, a leucite phase (KAlSi_2O_6 -JCPDS 38–1423) was formed first and then a phlogopite one ($\text{KMg}_3\text{AlSi}_3\text{O}_{10}\text{F}_2$ -JCPDS 16–0344). These phases are clearly identified after prolonged heating time ($t > 10$ h). During heating at the same temperature for a time $t = 20$ h, the formation of diopside was not detected. For the samples crystallized at

Table 1
Composition of the glass

Glass	Composition (mol%)								
	SiO_2	Al_2O_3	CaO	CaF_2	MgO	K_2O	Na_2O	Σ	F
Initial	53.5	11.5	12	–	16.5	6.5	–	100	
G-3	53.8 $\pm$ 0.5	12.3 $\pm$ 0.5	–	4.2 $\pm$ 0.1	20.3 $\pm$ 0.5	9.3 $\pm$ 0.1	0.1 $\pm$ 0.05	100	2 $\pm$ 0.1

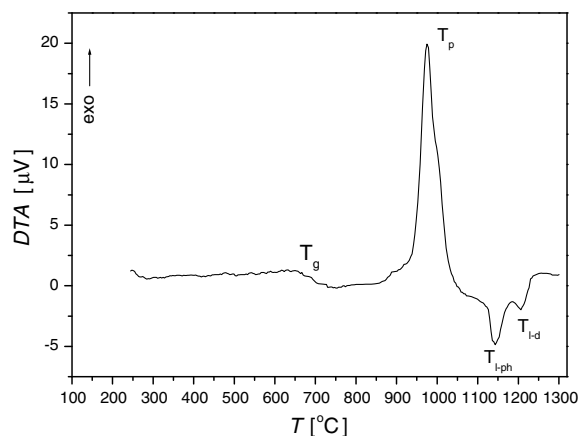


Fig. 1. DTA traces recorded at a heating rate of 10 °C/min up to $T = 1300$ °C, for powder sample particle size 0.4–0.63 mm.

Table 2

List of heat treatments and results of XRD analysis

T_n (°C)	t_n (h)	T_c (°C)	t_c (h)	Phase composition
		590, 630, 680	1–20	G
590	20	900	0.5–20	L, Ph, G
		950		L, Ph, D, G
		1060	20	L, Ph, D
630	20	900	0.5–20	L, Ph, G
		950		L, Ph, D, G
		990		L, Ph, D, G
		1060	20	L, Ph, D
		1100		
680	20	900	0.5–20	L, Ph, G
		950		L, Ph, D, G
		1000		L, Ph, D, G
		1060	20	L, Ph, D
		1185	1–3	L, Ph, D, G
		1250		L, Ph, D, G
		1300		L, Ph, G
		1350		L, Ph, G
		1400		L, G
		1420		L, G
		1450		L, G
		1485		G

L – leucite; Ph – phlogopite; D – diopside; G – glass.

$T_c = 950$ °C for shorter times ($t < 10$ h), the formation of leucite and phlogopite was also detected. With longer heating times ($t > 10$ h), diopside ($\text{CaMgSi}_2\text{O}_6$ -JCPDS 41–1370) as a third phase was detected. At crystallization temperatures $T_c = 1000$, 1060 and 1100 °C, the presence of all three phases was identified for nucleation temperatures $T_n = 590$, 630 and 680 °C. The presence of a glassy phase was not detected for the samples heated for $t_c = 20$ h at $T_c = 1060$ and 1100 °C. The XRD patterns of the samples nucleated at $T_n = 630$ °C and then crystallized at different temperatures are shown in Fig. 2.

To determine quantitatively the ratio of the crystalline phases, a sample was heated in the first step at $T_n = 630$ °C for $t_n = 20$ h and then at $T_c = 1100$ °C for $t_c = 50$ h. The XRD pattern of this sample was recorded in the 2θ range 5–100° with a step of 0.02° and a time of 4 s. A Rietveld profile matching mode fitting plot of this sample is presented in Fig. 3, while the results of the fitting are shown in Table 3. The obtained results show that the glass G-3 crystallized into a three component glass–ceramic, with leucite as the primary phase, and phlogopite and diopside as the secondary ones. The calculated unit-cell dimensions were compared with literature data and no divergence was registered.

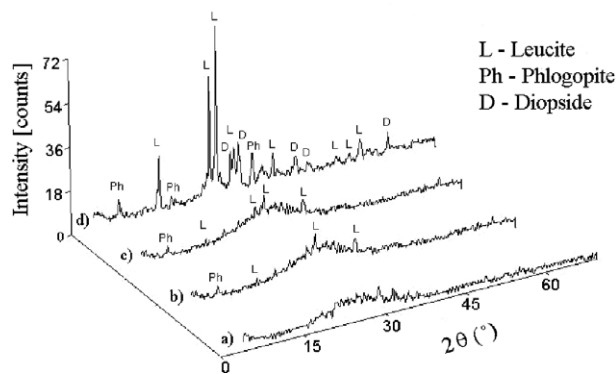


Fig. 2. XRD patterns of: (a) glass, (b) sample crystallized at $T_n = 590$ °C for $t = 20$ h and $T_c = 900$ °C for $t = 1$ h, (c) sample crystallized at $T_n = 630$ °C for $t = 20$ h and $T_c = 950$ °C for $t = 3$ h and (d) sample crystallized at $T_n = 630$ °C for $t = 20$ h and $T_c = 1100$ °C for $t = 20$ h.

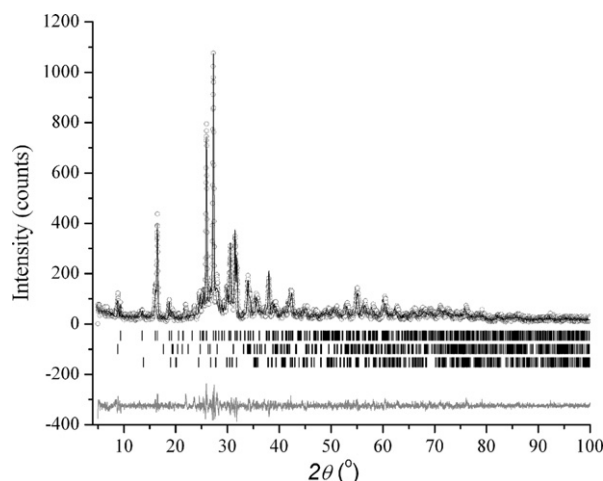


Fig. 3. Rietveld refinement plot of glass sample heat-treated at: $T_n = 630$ °C for $t = 20$ h and $T_c = 1100$ °C for $t = 50$ h. Observed (o), calculated (—) and difference powder diffraction profiles. Vertical bars mark Bragg peak positions.

A third series of experiment was performed by heating the samples in a one step regime at temperatures $T = 1185$ – 1485 °C. The results of XRD analyses show that at $T_c = 1185$ and 1250 °C, all three crystalline phases and a glassy phase were present. At $T_c = 1300$ and 1350 °C, leucite, phlogopite and a glassy phase appeared. At $T_c = 1400$ – 1450 °C, leucite and a glassy phase were detected. On increasing T_c , the intensity of the leucite phase decreased. The XRD patterns recorded on samples annealed at $T_c = 1185$, 1300 and 1420 °C are presented in Fig. 4.

To check the influence of the particle size of powder samples on the formation of the phases under non-isothermal conditions, experiments were performed with samples of particle size 0–1 mm, which were heated at a rate of $v = 10$ °C/min up to the crystallization peak temperatures. The XRD analyses show that all three crystalline phase were formed in all samples, indicating that the particle size of the glass does not influence the phase composition of the crystallized samples, Fig. 5.

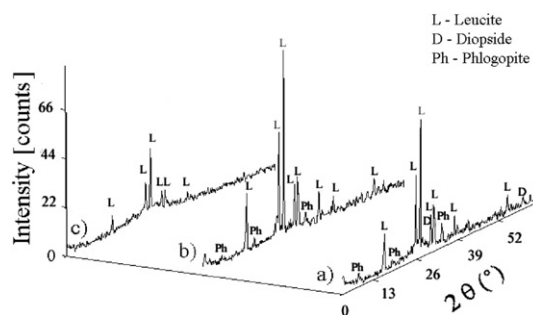
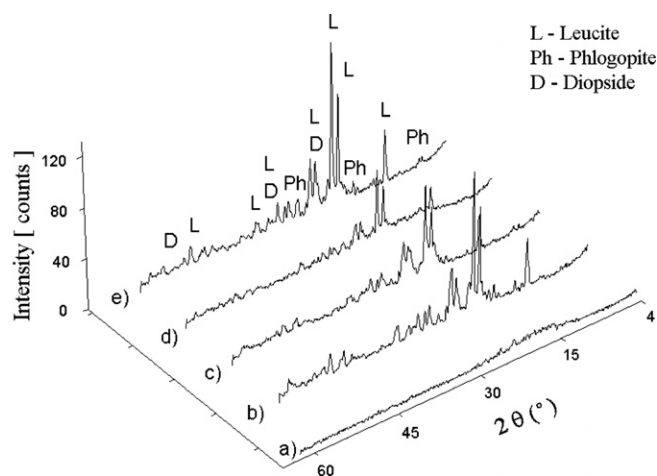
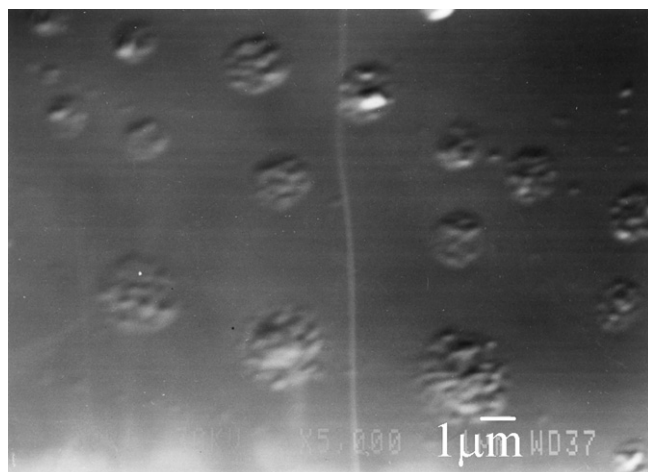
3.4. Microstructural analysis

The SEM microphotograph of the interior surface of a nucleated glass annealed at $T_n = 630$ °C for $t_n = 20$ h is presented in Fig. 6, in which a droplet region is clearly visible. XRD analysis of the nucleated samples (Table 2) did not show any traces of crystallites (some Ca or Mg fluorides). This indicates that during heating under

Table 3

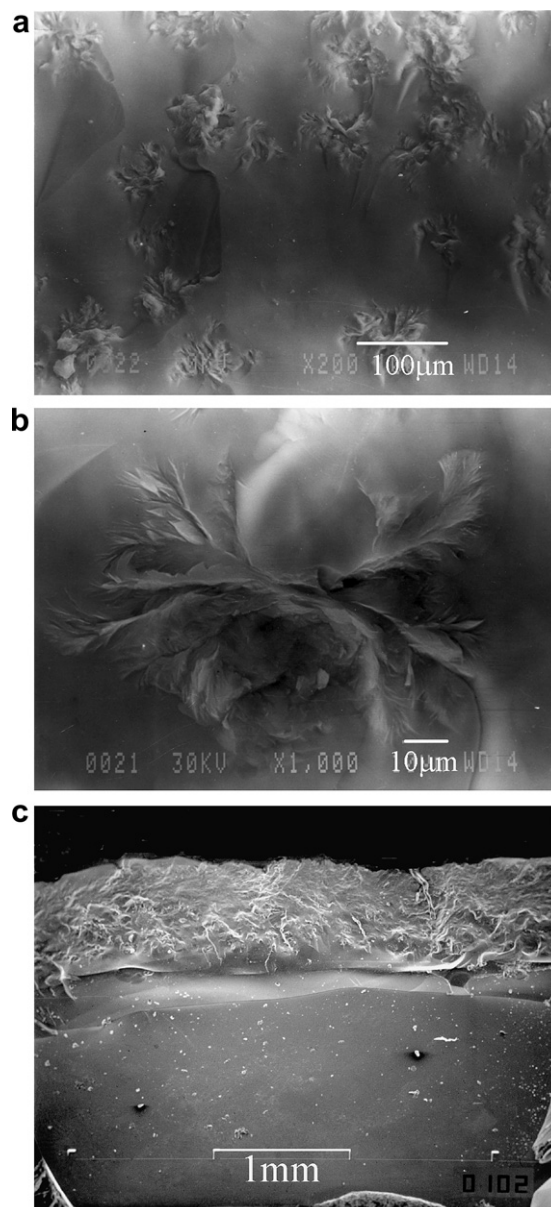
The most important crystallographic parameters for crystalline phases, obtained from Rietveld refinement of XRD pattern

Phase	Unit-cell parameters				Reliability factors		Quantitative volume fraction (%)
	<i>a</i> (Å)	<i>b</i> (Å)	<i>c</i> (Å)	β (°)	<i>R</i> _B (%)	<i>R</i> _F (%)	
Leucite	13.0765(9)	13.0765(9)	13.739(1)	90	1.13	1.41	68(1)
Phlogopite	5.325(4)	9.201(7)	10.138(9)	98.31(7)	7.65	5.27	19(1)
Diopside	9.739(5)	8.887(4)	5.259(2)	106.12(3)	2.78	2.78	13(1)

**Fig. 4.** XRD patterns of: samples annealed at: (a) $T_c = 1185$ °C for $t_c = 1$ h, (b) $T_c = 1300$ °C for $t_c = 3$ h and (c) $T_c = 1420$ °C for $t_c = 3$ h.**Fig. 5.** XRD patterns of (a) the initial glass and of powder samples of particle size: (b) <0.038 mm heated up to $T = 1007$ °C, (c) 0.038–0.53 mm heated up to $T = 1036$ °C, (d) 0.106–0.212 mm heated up to $T = 984$ °C and (e) 0.63–0.701 mm heated up to $T = 991$ °C. Heating rate was $\nu = 10$ °C/min.**Fig. 6.** SEM micrographs for glass samples annealed at: $T_n = 630$ °C for $t_n = 20$ h.

these conditions, phase separation precedes the homogenous nucleation.

The evolution of the microstructure is presented in Figs. 7 and 8. In Fig. 7(a) and (b), SEM microphotographs of the samples thermally treated at $T_n = 630$ °C for $t_n = 20$ h and then at $T_c = 950$ °C for $t_c = 3$ h are shown. Crystal growth in the shape of two-dimensional equiaxial dendrites with a pronounced anisotropy along the direction $\langle 001 \rangle$, typical for leucite, is visible. The expressed anisotropy indicates a layered, diffusion controlled growth, proceeding on a

**Fig. 7.** SEM micrographs for glass samples annealed at: (a) and (b) $T_n = 630$ °C for $t_n = 20$ h and $T_c = 950$ °C for $t_c = 3$ h and (c) $T_c = 1420$ °C for $t_c = 3$ h.

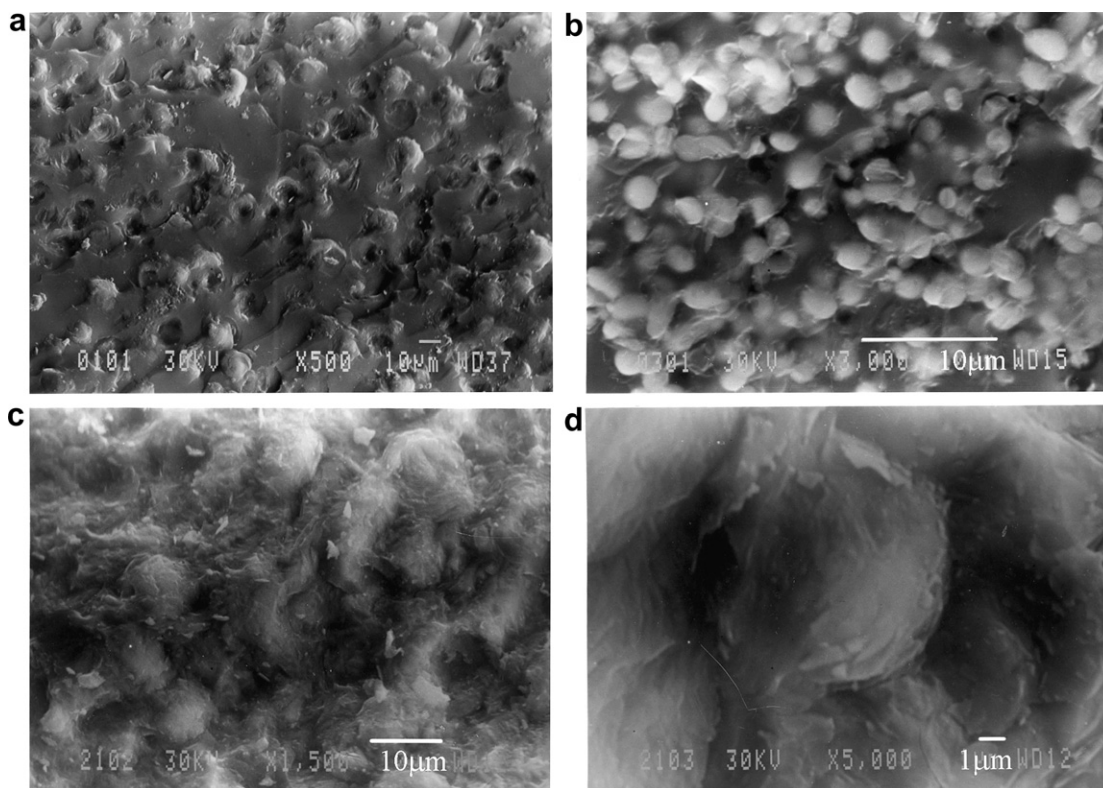


Fig. 8. SEM micrographs for glass samples annealed at: (a) $T_n = 590$ °C for $t_n = 20$ h and $T_c = 900$ °C for $t_c = 1$ h and (b) $T_n = 590$ °C for $t_n = 20$ h and $T_c = 900$ °C for $t_c = 2$ h and (c) and (d) $T_n = 630$ °C for $t_n = 20$ h and $T_c = 1060$ °C for $t_c = 2$ h.

faceted, atomically smooth interface. The XRD pattern of this sample Fig. 2(c) shows that leucite with traces of phlogopite was formed. The crystallized samples, even after short times of thermal treatment, became fully opaque (white), so that it was not possible for the crystallization mechanism of the leucite phase to be defined (surface or volume). By increasing the crystallization temperature above $T_c > 1300$ °C, the sample opacity decreased. XRD measurements showed that the amount of crystalline phases also decreased and that a diopside phase was not present (Fig. 4). At temperatures above $T_c > 1400$ °C, the samples were not opaque and had a glassy phase in the center and a white crystalline layer on the sample surface became visible. On further increasing the temperature, the thickness of crystalline layer decreased and the amount of glassy phase increased. The XRD patterns of these samples showed the presence of leucite and a glassy phase (Table 2 and Fig. 4). The SEM microphotograph of the sample crystallized at $T_c = 1420$ °C for $t = 3$ h is presented in Fig. 7(c), which shows a white crystalline surface layer and a glassy phase. At $T_c = 1485$ °C after annealing the sample for $t = 3$ h, the crystallized layer had disappeared and only a glassy phase was present. These results show that the leucite phase in this glass was formed by surface crystallization.

The SEM microphotograph of the interior surface of a sample thermally treated in a two-stage regime at $T_n = 590$ °C for $t_n = 20$ h and then at $T_c = 900$ °C for $t_c = 1$ h is shown in Fig. 8(a). Spherical-shaped, white crystals of about 1 µm in diameter are visible. The XRD pattern of this sample is presented in Fig. 2(b), showing that in addition to leucite, a phlogopite phase was formed. The SEM microphotograph of the interior surface of a sample thermally treated at $T_c = 900$ °C for $t_c = 2$ h is shown in Fig. 8(b). A large number of spherical-shaped crystals with diameters of about 2 µm were present in the glass. The SEM microphotographs of the interior surface of a sample thermally treated at $T_n = 630$ °C for

$t_n = 20$ h and $T_c = 1060$ °C for $t_c = 2$ h are shown in Figs. 8(c) and (d), which illustrate the growth of these crystals at the higher temperature. Their dimensions were 10 µm and the growth morphology remained unchanged. These results indicate that the phlogopite phase in this glass is formed by direct volume crystallization, without the formation of precursor crystal phases with spherulite crystal growth morphology.

Diopside should grow with dendrite morphology as rod-like crystals by the surface crystallization mechanism when the glass is reheated [10–12]. However, such crystals were not detected. The reason for this is that the diopside phase was formed last in the smallest volume (Tables 2 and 3). Additionally, the investigation was performed with samples containing a high amount of glassy phase (initial stage of glass crystallization), which enabled the crystal growth morphology to become clearly visible. Therefore, the number of diopside crystals was small.

3.5. Mechanism of crystallization of glass powder

Differential thermal analysis is a convenient and fast procedure for evaluating the dominant crystallization mechanism of glass powder [13]. Experimental and theoretical studies have shown that the particle size of glass powder influences the mechanism of its crystallization [14,15]. In the case of polymorphic crystallization by volume and surface nucleation and interface-limited growth, the height of the exothermal DTA peak (δT_p) is proportional to total number of nuclei (volume and surface) contained in the glass particle. Accordingly, the DTA peak height is sensitive to the contributions of these nuclei in glass particles. If surface and volume crystallization proceed simultaneously, three distinct regions can be distinguished as a function of particle size. They reflect behavior arising from a decreasing surface-to-volume ratio with increasing particle size. The ratio $T_p^2/(\Delta T)_p$, where T_p is the

DTA peak temperature and $(\Delta T)_p$ is the half-width of the DTA peak, shows a similar dependency on particle size. This ratio is related to the dimension of crystal growth and can be used as a qualitative measure for surface and volume crystallization.

To determine the dominant crystallization mechanism of this glass, DTA curves of glass powder samples of particle sizes 0–1 mm were recorded at a heating rate of $v = 10^\circ\text{C}/\text{min}$. All the DTA curves showed only one exothermal peak, appearing in temperature range $T_p = 967\text{--}1036^\circ\text{C}$. Additionally, dilatometric measurements indicated that this glass has a transformation temperature at $T_g = 680^\circ\text{C}$. Based on these results, it may be concluded that the temperature interval of nucleation and crystal growth vary considerably. Therefore, the choice of heating rate enables a certain number of nuclei to be formed in the glass transformation region, where the nucleation rate is significant and the crystal growth rate is negligible. This number of nuclei does not significantly change with further heating in the higher temperature interval because the nucleation rate in this range is low, while the crystal growth rate becomes considerable. Therefore, crystal growth in this glass proceeds with a constant number of nuclei under non-isothermal conditions in the DTA run.

The effect of particle size on the exothermal DTA peak temperature, T_p , is presented in Fig. 9. It may be seen that curve (g) has three regions. For a particle size range of 0–0.25 mm, T_p decreased, increased for 0.25–0.5 mm particles and then remained constant for the particle size range of 0.5–1 mm.

The results of the influence of particle size on the DTA peak height $(\delta T)_p$ and $T_p^2/(\Delta T)_p$ ratio are shown in Fig. 10. To show clearly this dependency, the dotted curve in Fig. 10 is the average value of both parameters.

3.6. Nucleation behavior

A rapid and convenient method based on DTA measurements has been suggested for the determination of nucleation parameters in glasses [16,17]. In this method, a small amount of glass sample is first heated isothermally (nucleated) at a temperature (T) for a given time (t) and then followed by a DTA scan at a constant heating rate until complete crystallization (as indicated by the exothermic DTA peak). The measurements were repeated with different values of T , while t , the heating rate v , sample weight and particle size were kept constant. A plot of the maximum DTA peak height $(\delta T)_p$, or the inverse peak temperature (T_p^{-1}), as a function of the nucleation temperature gave a nucleation-rate-like curve which, to a reasonable degree of accuracy, determined the temperature range for nucleation and the temperature of the maximum nucleation

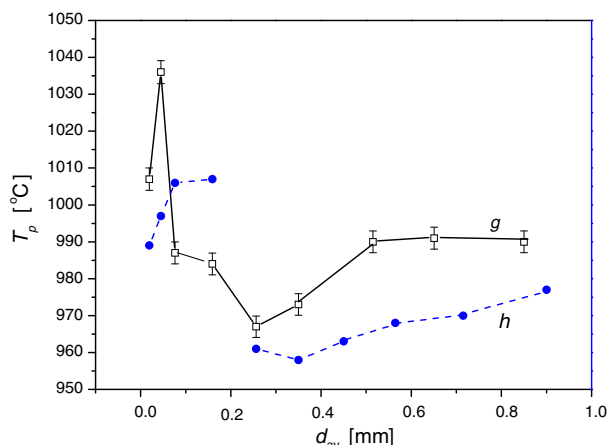


Fig. 9. Effect of particle size on DTA exothermal peak temperature T_p : (g) this glass and (h) glass with 1.5 mol% of F⁻ anions [4]. Lines are drawn as guides to the eyes.

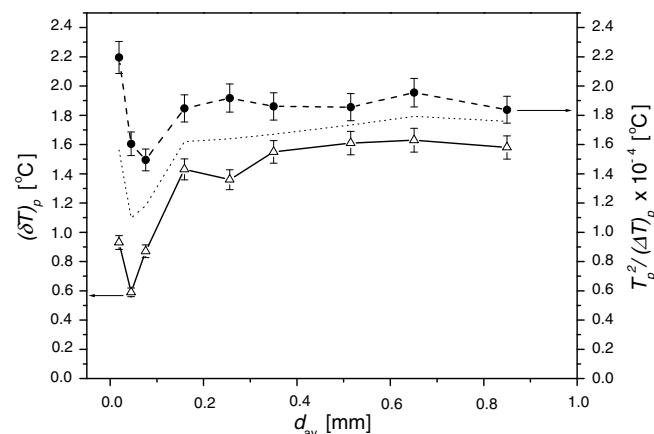


Fig. 10. Effect of particle size on the DTA exothermal peak height $(\delta T)_p$ and the ratio $T_p^2/(\Delta T)_p$. Lines are drawn connecting data symbols of each kind as guides to the eyes. General trend (.....).

rate. The difference between the peak temperature of the as-quenched glass T_{pq} and the peak temperature of previously heat-treated glass (T_{pt}) ($\Delta T_p = T_{pq} - T_{pt}$) also showed a similar behavior.

The method was demonstrated on glasses which crystallize polymorphically by homogeneous nucleation [16–20]. Theoretical considerations showed that within the framework of the classical theory of nucleation, it is reasonable to expect such maximums in these quantities [18,21]. In order to apply this method, the following conditions must be fulfilled: (a) the nucleation is completed prior to crystal growth, i.e., the crystal growth occurs on a constant number of nuclei during the non-isothermal treatment and (b) volume nucleation should occur in the glass.

Recently the results of the shifts of the DTA peak in a glass undergoing primary crystallization were reported [22]. As in the case of primary crystallization, the compositions of the crystal phase were different from those of the glass, the equilibrium fraction of crystals was temperature dependent and can be much less than 100%. Also, the crystal growth rate can be controlled by diffusion. Therefore, Johnson–Mehl–Avrami–Kolmogorov (JMAK) formalism-based models cannot be used to describe the kinetics of primary crystallization [23].

As shown in Section 3.5, the nucleation and crystal growth in the presently studied glass proceed in temperature ranges which differ by more than 200°C . Therefore, the corresponding choice of heating rate during the DTA experiment facilitates the satisfaction of condition (a). Also, the results presented in Section 3.5 show that in powder samples of average particle size of >0.2 mm volume crystallization is dominant, thus satisfying condition (b). Consequently, a glass powder with particle sizes between 0.4–0.63 mm and a heating rate of $v = 10^\circ\text{C}/\text{min}$ were chosen for these experiments. The effect of the pre-DTA heat treatment time at fixed temperatures of 560, 610, 630, 660 and 680°C on T_p , $(\delta T)_p$ and ΔT_p are presented in Fig. 11.

A second group of experiments was performed on samples heated for a fixed pre-DTA heating times of $t = 1, 3$ and 5 h at temperatures in the range of $T = 550\text{--}750^\circ\text{C}$. The pre-DTA heat treatment times were chosen based on the results shown in Fig. 11. The curves of the change of the parameters T_p , $(\delta T)_p$, and ΔT_p , with temperature for fixed times are presented in Fig. 12.

4. Discussion

The influence of the average values of the particle size on the DTA peak temperature T_p is shown in Fig. 9. This curve differs from the curve obtained for glasses crystallized polymorphically, where

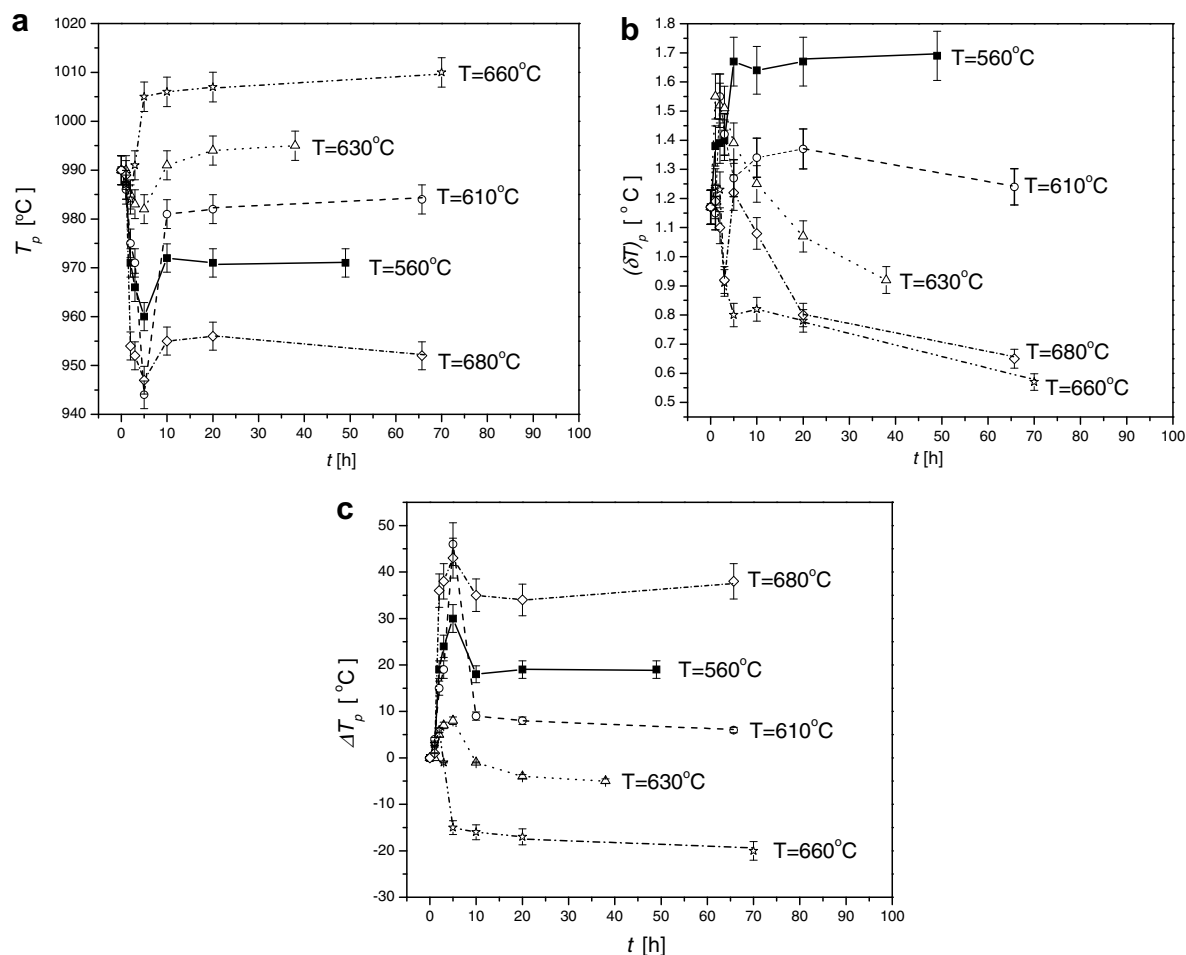


Fig. 11. Effect of the pre-DTA heat treatment time at the temperatures in range 550–750 °C, on: (a) T_p , (b) (δT_p) , and (c) ΔT_p . Lines are drawn connecting data symbols of each kind as guides to the eyes.

T_p increased with increasing particle size, regardless of the dominant crystallization mechanism [15]. In order to understand such a behavior, curve *h*, referring to a glass of similar composition containing 1.5 mol% F^- anions, which crystallizes with the same phase composition but with different amounts of the phases in the glass, is also presented in Fig. 9 [4]. As may be seen from curve *h*, increasing the particle size led to a remarkable change in the value of T_p . The result of this investigation showed that for samples of average particle size <0.15 mm, leucite and diopside phases were formed and the corresponding crystallization peaks occurred at higher temperatures. For a particle size about 0.15 mm, the formation of a phlogopite phase was also evidenced and for samples of average particle size >0.15 mm, all three phases, leucite, diopside and phlogopite, were present, with the corresponding crystallization peaks in the lower temperature range.

To explain the behavior of glass G-3 (curve *g*), additional experiments were performed in which the powder samples were heated at a rate $v = 10$ °C/min until the temperature of the crystallization peaks was attained. XRD analyses showed that the phase composition of the crystallized samples did not change for samples of different particle size (Fig. 5). This suggests that the content of F^- anions in glass G-3 enabled the formation of a phlogopite phase in samples having the smallest particle sizes, where surface crystallization dominates. As different phases appeared during primary crystallization, their contributions to the DTA signal were also different. With increasing particle size, the number of surface nuclei decreased. As it was shown that the leucite and diopside

phases were formed by the surface mechanism, their quantitative amounts consequently decreased as the particle size of the glass increased. On the other hand, the phlogopite phase is formed via both the surface and volume mechanism. Thus, as increasing the particle size leads to a decrease in the number of surface nuclei and a simultaneous increase in the number of volume nuclei, the total number of phlogopite nuclei increases continuously resulting in an increase of the quantitative amount of this phase and its contribution to the DTA signal. As the phlogopite phase is formed at lower temperatures than the primary phase, increasing the quantitative amount of phlogopite phase caused a decrease in the DTA crystallization peak temperature to $T = 1036$ – 967 °C. When the particle sizes were in the range of 0.2–0.3 mm, the number of phlogopite volume nuclei formed became equal to the sum of the surface nuclei corresponding to all three phases and the crystallization peak temperature reached its minimum value. On further increasing the particle size to >0.3 mm, the number of surface nuclei of leucite and diopside decreased and, consequently, also their amounts, while the number of volume phlogopite nuclei increased and also its content, until the quantitative ratio of the phases, which is determined by the glass composition, was attained. In this particle size range, the crystallization peak temperature increased to $T = 967$ – 990 °C. These results showed that in the case of primary crystallization, in addition to particle size, the occurrence of different crystallization mechanism significantly influenced the changes in the DTA crystallization peak.

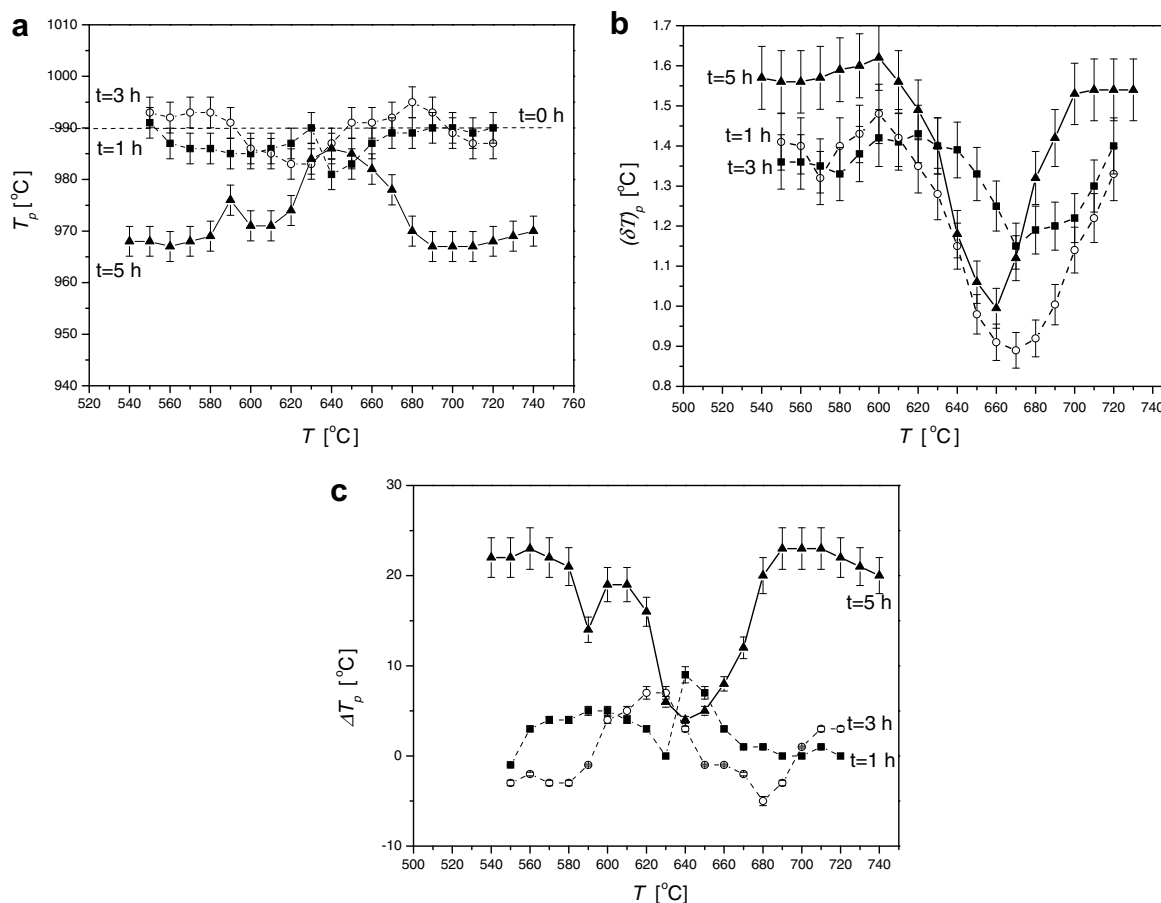


Fig. 12. Effect of the nucleating temperature on: (a) T_p , (b) $(\delta T)_p$ and (c) ΔT_p . Lines are drawn connecting data symbols of each kind as guides to the eyes.

The influence of particle size on the parameters $(\delta T)_p$ and $T_p^2/(\Delta T)_p$ is shown in Fig. 10. It may be seen that similar curves having three regions were obtained for both parameters. In the grain size range of 0–0.1 mm (smallest particle sizes, region 1), both parameters decreased with increasing particle size and surface crystallization dominated. Both surface and volume crystallization were important in the size interval 0.1–0.2 mm (region 2), which immediately followed the minimum and extended to the point where the curve nearly reached its asymptotic value. Volume nucleation and growth was significant for the large particle sizes of 0.2–1 mm (region 3). This dependency is similar to that of polymorphic crystallization and interface-limited growth, but the widths of all regions were narrow. Region 1 of this glass is in the grain size range of 0–0.1 mm, whereas for polymorphic crystallization, a size range of 0–0.4 mm was anticipated [15]. Region 2 of this glass is in the size range of 0.1–0.2 mm, whereas for polymorphic crystallization, a range of 0.6–4.5 mm was anticipated [15]. Region 3 of this glass is in the grain size range of 0.2–1 mm, whereas for polymorphic crystallization, the size interval is >4.5 mm [15]. Also, the ratio of the minimum and maximum on the curves for the parameter $T_p^2/(\Delta T)_p$ is 1.3 and parameter $(\delta T)_p$ is 2.7. In the case of polymorphic crystallization, this ratio is anticipated to be 3. Such behavior is also influenced by the simultaneous action of two crystallization mechanisms.

The results of the study of the nucleation behavior are shown in Figs. 11 and 12. It may be seen in Fig. 11(a) that T_p decreased up to $t = 5$ h, then increased to a value which was close to that obtained with an as-quenched sample and then remained constant. This curve differs from the curve obtained for polymorphically crystallized glasses, when T_p shifts to lower temperatures with increasing

time of heat treatment. Also, this difference exists regarding the curve obtained for a primary crystallized glass when the primary phase was nucleated in bulk. In this case, T_p decreased initially but increased later on, becoming significantly larger than the initial value [22]. The first part of the curve, Fig. 11(a), may be explained, as in the case of polymorphic crystallization, by an increase in the number of nuclei (phlogopite nuclei) with increasing heat treatment time up to $t = 5$ h, causing a reduction in the exothermal peak temperature T_p . After time $t = 5$ h, new nuclei of the phlogopite phase were not formed, hence the sum of the nuclei remained constant and the exothermal peak temperature did not change with increasing treatment time. A similar change of the parameter ΔT_p occurred, Fig. 11c. The changes of the exothermal peak height, $(\delta T)_p$, as a function of time at the chosen temperatures are presented in Fig. 11(b). These curves are not completely similar, as is the case for the parameters T_p and ΔT_p . The maximum of the curves appeared at shorter times $t < 5$ h. On increasing the time of heat treatment, the curves declined with different slopes. Based on these results, it may be concluded that the decrease of T_p and the increase of $(\delta T)_p$ and ΔT_p in the pre-DTA heat treatment time of $t \leq 5$ h were induced by changes in the number of nuclei of the secondary phase phlogopite.

The changes of these DTA parameters with temperature at different fixed times are shown in Fig. 12, from which it can be seen that complex curves were obtained. On increasing the annealing time, the changes in the curves become remarkable. They were large for an annealing time of $t = 5$ h. The changes in T_p with nucleation temperature during heating of the sample for different times are shown in Fig. 12(a), in which the T_p of the as-quenched glass sample (dashed line $t = 0$ h) is also shown. It can be seen that the

T_p of the previously heat-treated glass is usually lower than the T_p of the as-quenched glass.

In temperature range of $T = 580\text{--}660\text{ }^\circ\text{C}$, the changes matched the nucleation curve (bell shape), indicating the occurrence of the nucleation process of phlogopite. To analyze this possibility, the nucleation rate curve of phlogopite was estimated according to the equation for a steady-state nucleation rate [24,25]:

Table 4

Data used for the numerical calculation of homogenous nucleation rate I of phlogopite

$k = 1.380622 \cdot 10^{-3} \text{ J/K}$	$A = -3.1474$
$h = 6.626176 \cdot 10^{-34} \text{ Js}$	$B = 8255.3$
$\lambda = 5.88 \cdot 10^{-10} \text{ m}$	$T_o = 415.27$
$T_g = 953.15 \text{ K}$	$V_{m-ph} = 148.850^{-6} \text{ m}^3/\text{mol}$
$T_l = 1758.15 \text{ K}$	$\eta_o = 10^{-3.1474} \text{ Pa s}$
$T_{l-ph} = 1431.15 \text{ K}$	$\eta(T_g) = 10^{12.2} \text{ Pa s}$
$\Delta H_{m-ph} = 308779 \text{ J/mol}$	$R = 8.314 \text{ J/mol K}$
$N_A = 6.02210^{23} \text{ mol}^{-1}$	

* [26].

$$I = I_0 \exp \left[-C_{22} \left(\frac{T_{gr} - T_{or}}{T_r - T_{or}} \right) - \left(\frac{C_1}{T_r(1 - T_r)^2} \right) \right], \tag{1}$$

where

$$I_0 = (N_v \cdot k \cdot T)/h, \tag{2}$$

$$C_1 = (16\pi\alpha^3\Delta H_m)/(3RT_1), \tag{3}$$

$$C_{22} = \ln[\eta(T_g)/\eta_o], \tag{4}$$

$$T_r = T/T_1, \quad T_{or} = T_o/T_1, \quad T_{gr} = T_g/T_1, \tag{5}$$

where $N_v \cong 1/\lambda^3$ is the number of structural units of size λ per unit volume of glass, k and h are the Boltzmann and Plank constant, respectively, T is the temperature, α is an empirical constant, ΔH_m is the enthalpy of melting of the crystalline phase, R is the gas molar constant. The values of T_r , T_{or} and T_{gr} were calculated for the liquidus temperature of phlogopite (T_{l-ph}). The values of parameters employed for the calculations are given in Table 4. For the calculation the parameter C_1 (the thermodynamic barrier for nucleation), according to Eq. (3), the value of the empirical constant α has to be defined. Under the assumption that the crystal/melt surface

Table 5

The values of C_1 and I as function of α at $T = 630\text{ }^\circ\text{C}$

	α					
	0.1	0.2	0.3	0.4	0.5	0.55
C_1	0.44	3.48	11.74	27.83	54.35	72.34
$I \text{ (m}^{-3} \text{ s}^{-1}\text{)}$	$5.89 \cdot 10^{21}$	$2.01 \cdot 10^6$	$2.29 \cdot 10^{-36}$	$4.55 \cdot 10^{-118}$	$9.88 \cdot 10^{-253}$	$4.43 \cdot 10^{-344}$

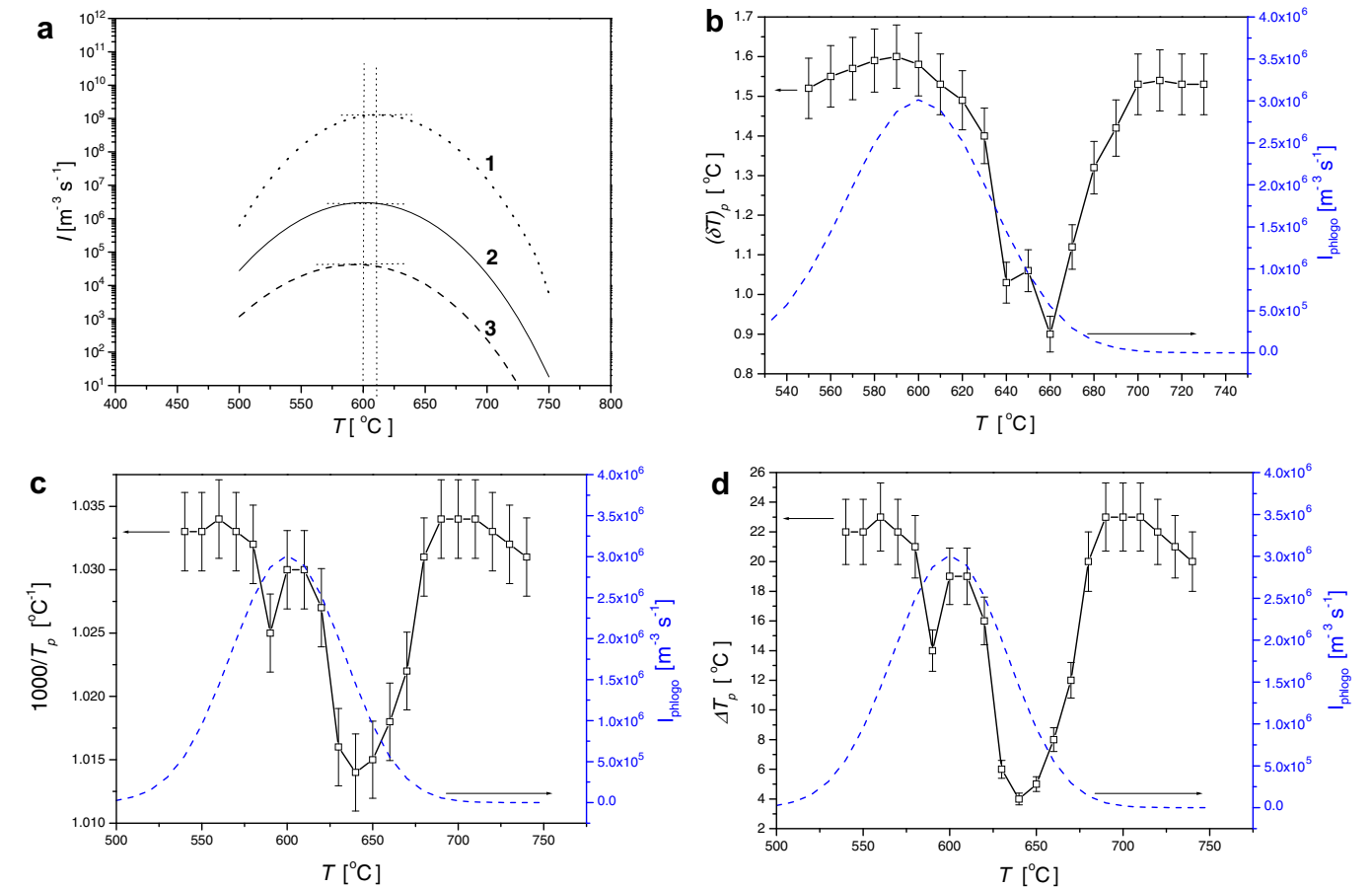


Fig. 13. The nucleation rate curves of phlogopite: (a) curve 1– I for $\log \eta = 4$ at T_l , curve 2– I for $\log \eta = 3$ at T_l and curve 3– I for $\log \eta = 2.4$ at T_l . The curves were calculated according to Eq. (1), (b) $(\delta T)_p$ as function of temperature for annealing time $t = 5$ h and calculated curve I using Eq. (1), (c) $1000/T_p$ as function of temperature for annealing time $t = 5$ h and calculated curve I using Eq. (1) and (d) ΔT_p as function of temperature for annealing time $t = 5$ h and calculated curve I using Eq. (1).

energy (σ) is temperature and size independent, α is obtained from fitting the nucleation experiments and varies from 0.4–0.55 for several silicate glasses which crystallized polymorphically [24,27,28]. However, using these values in Eqs. (1) and (3), a low nucleation rate and high σ values for the phlogopite phase were obtained. Accordingly, the influence of the value of α over a wide range was examined. The values of C_1 and I as function of $\alpha = 0.1$ –0.55 at $T = 630^\circ\text{C}$ are given in Table 5, from which it may be seen that for $\alpha < 0.2$, the obtained values are high and for $\alpha > 0.2$ too low. Using $\alpha = 0.2$ and the Scapski–Turnbull equation [27]:

$$\sigma = \alpha \Delta H_{\text{m-ph}} V_{\text{m-ph}}^{-2/3} N_A^{-1/3}, \quad (6)$$

where N_A is the Avogadro number, a value of the surface tension of the phlogopite of $\sigma = 0.266 \text{ J m}^{-2}$ was obtained.

The parameter $C_{22}(T_{\text{gr}} - T_{\text{or}})$ in first term of Eq. (1) corresponds to the kinetic barrier of nucleation. For the calculation of C_{22} according to Eq. (4), the viscosity curve has to be known. Since, the results in Section 3.3 indicate that the leucite phase is dominant, the liquidus temperature of leucite was assumed as the glass liquidus temperature in this analysis. Also, the determination of the liquidus temperature at $T = 1485^\circ\text{C}$ and the cast experiment of this glass at $T = 1530^\circ\text{C}$ showed that this glass belongs to “long” glasses. To investigate the influence of the choice of $\log \eta$ at T_i on the nucleation rate curve of phlogopite, the calculations were performed for $\log \eta$ (Pa s) = 2.4, 3 and 4 at T_i , and the results are given in Fig. 13(a). It may be seen from this figure that the nucleation rate curves do not change position for viscosity changes at T_i , but their magnitude changes. A comparison with the experimental results was performed for a nucleation rate estimated for $\log \eta$ (Pa s) = 3 at T_i . In this case, the Vogel–Fulcher–Tammann (VFT) equation was fitted to describe the viscosity of this glass as

$$\log \eta (\text{Pa s}) = -3.1474 + 8255.3 / (T - 415.27), \quad T [\text{K}]. \quad (7)$$

In Fig. 13(b)–(d), the experimental curves $(\delta T)_p$, $1000/T_p$ and ΔT_p for an annealing time of 5 h are compared with the calculated nucleation rate curve of phlogopite. It may be seen that the changes in the parameter curves appeared in the same temperature range as in the curve of the phlogopite nucleation rate. The maximums of these curves are compatible with the maximums of the calculated nucleation rate curve.

As the nucleation occurs in the high viscosity region of this glass, transient nucleation is present. The time-lag (τ) for phlogopite nucleation can be estimated by application of the following equation [24]:

$$\tau = \frac{16hN_A}{\pi^2 \Delta H_m} \frac{\alpha}{(1 - T_r)^2} \exp \left[C_{22} \left(\frac{T_{\text{gr}} - T_{\text{or}}}{T_r - T_{\text{or}}} \right) \right]. \quad (8)$$

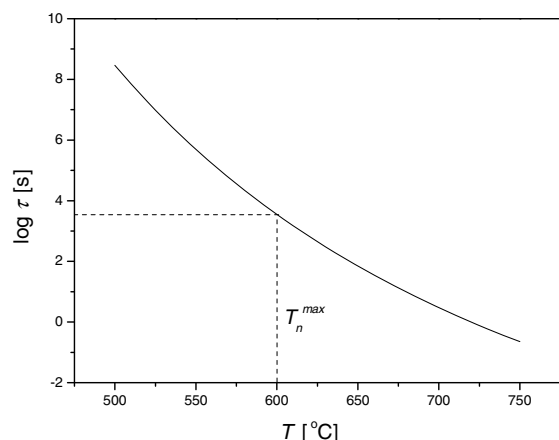


Fig. 14. The calculated curve of $\log \tau$ vs. T .

Employing the values of the parameters given in Table 4, the time-lag τ was calculated. In Fig. 14 the calculated curve of $\log \tau$ vs. T is shown. A transient time of $\tau = 3531 \text{ s}$ ($t_{\text{ind}} = (\pi^2/6)\tau = 1.61 \text{ h}$) at $T_n^{\text{max}} = 600^\circ\text{C}$ was determined. The experimental curve recorded (Fig. 12) shows that T_n^{max} is not clearly visible for a pre-DTA heat treatment time of $t = 1 \text{ h}$. This could be explained if the steady-state nucleation rate at T_n^{max} had not yet been attained with this treatment time and, thus, that the number of nuclei present in the glass was insufficient to subsequently influence the DTA signal. For pre-DTA heat treatment times $t = 3$ and 5 h , T_n^{max} are visible on the curves because the times were sufficiently long for a steady-state rate of nucleation.

5. Conclusions

In this study, the crystallization of a mixed anion silicate glass containing 2 mol% of F^- anions was investigated using DTA, XRD and SEM methods. This glass crystallizes with primary crystallization. The primary phase leucite and the secondary phase diopside were formed by surface crystallization, with dendrite crystal growth morphology, and the secondary phase phlogopite was formed by volume crystallization, with spherulite growth.

Analysis of the dominant crystallization mechanism of a powder glass sample showed that the behavior of the DTA peak temperature, T_p , as a function of particle size is complex. This behavior of T_p is different to that observed in the case of polymorphic crystallization. In addition to particle size, the quantitative ratio of the different phases which were formed also influences the changes in T_p . The behaviors of the parameters $(\delta T)_p$ and $T_p^2/(\Delta T)_p$ depend only on the change in the surface-to-volume nuclei ratio, as is the case with polymorphic crystallization. However, the particle size ranges where the surface, volume and mixed crystallization mechanism dominate are considerably narrower for the studied glass.

In this glass, the nucleation and crystal growth proceed in temperature ranges that differ by more than 200°C . This enabled the DTA method to be applied for a study of the nucleation. The T_p of the samples thermally treated for different times, at fixed temperatures, decreased up to $t = 5 \text{ h}$ and then increased to a value close to that of an as-quenched sample, after which it remained constant. The curves of $(\delta T)_p$ and ΔT_p as a function of time for fixed temperatures also show a maximum. These results are different from those previously reported for the case of polymorphic crystallization. Such behavior is influenced by the change in the number nuclei of the secondary phase phlogopite.

The changes in T_p and $(\delta T)_p$ and ΔT_p as a function of temperature for fixed times were also complex curves. On increasing the pre-DTA heat treatment time, the changes of the curves become significant. In the temperature range 580 – 660°C , the changes in the curves were similar to those observed in the nucleation curve, which indicates that the nucleation process proceeded in this temperature range. The temperature of the maximum nucleation rate was $T_n^{\text{max}} = 600^\circ\text{C}$. Analysis of the nucleation of phlogopite according to the classical nucleation theory gave a nucleation curve in same temperature interval. The induction time estimated at T_n^{max} was 1.61 h . This shows that the changes in the experimental curves recorded for fixed times of 3 and 5 h were clearly visible because the steady-state nucleation rate had been attained.

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